# VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

**Anonymous Authors**
Institution
email@domain.com

## Abstract

While text-only models like GPT-o1 have made remarkable progress in mathematical and scientific reasoning through slow-thinking approaches, multimodal vision-language models (VLMs) continue to lag significantly in comparable reasoning tasks. This work addresses the challenge of developing robust, explicit self-reflection capabilities in VLMs through direct reinforcement learning, avoiding costly distillation pipelines. We identify that standard Group Relative Policy Optimization (GRPO) suffers from vanishing advantages as rewards saturate, which starves learning signals and discourages deeper reasoning in multimodal settings. To address this, we propose `VL-Rethinker`, which combines GRPO with two key innovations: (1) **Selective Sample Replay (SSR)**, which stores and prioritizes experiences with non-zero advantages to counter vanishing gradients, and (2) **Forced Rethinking**, which explicitly trains self-verification and correction capabilities by appending rethinking triggers during training. Our approach is trained on approximately 38.9K curated, verifiable multimodal queries and learns to rethink selectively rather than uniformly. `VL-Rethinker` achieves new state-of-the-art results across multiple benchmarks: our 72B model reaches 80.4% on MathVista testmini (+5.6 over open-source SoTA), 63.5% on MathVerse testmini (+6.3), and 44.9% on MathVision test (+6.8). Our 7B model also substantially outperforms existing reasoning VLMs, demonstrating the effectiveness of our approach across model scales.

## 1 Introduction

The emergence of "slow-thinking" models like OpenAI's GPT-o1 has demonstrated remarkable capabilities in mathematical and scientific reasoning by employing explicit reasoning chains and self-verification strategies. However, this progress has been largely confined to text-only domains, with multimodal vision-language models (VLMs) showing limited comparable gains in visual reasoning tasks. While VLMs have achieved impressive performance on basic visual understanding tasks, they still struggle with complex multimodal reasoning that requires deep, step-by-step analysis of visual and textual information together.

Current approaches rely on supervised fine-tuning with chain-of-thought data or costly distillation from proprietary models. These methods have limitations: supervised fine-tuning leads to imitation without understanding failure modes, while distillation requires access to teacher models and substantial resources.

Reinforcement learning (RL) offers a promising alternative for training models to develop genuine reasoning capabilities by learning from both positive and negative feedback. However, applying RL to vision-language reasoning presents unique challenges. Standard policy optimization methods like Proximal Policy Optimization (PPO) and Group Relative Policy Optimization (GRPO) face the

Preprint. Under review.

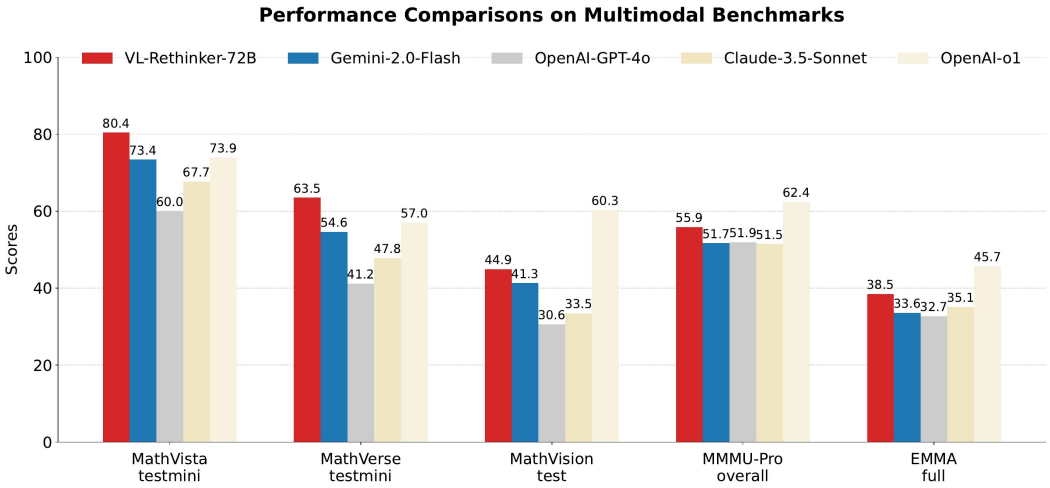


Figure 1: Performance comparison between `VL-Rethinker` and other state-of-the-art models across different multimodal reasoning benchmarks. Our approach achieves significant improvements across all tested benchmarks, demonstrating the effectiveness of combining Selective Sample Replay with Forced Rethinking for multimodal reasoning.

problem of *vanishing advantages* in multimodal settings, where rewards quickly saturate as models improve, leading to sparse learning signals that discourage deeper reasoning.

In this work, we introduce `VL-Rethinker`, a novel approach that addresses these challenges through two key innovations. First, we propose **Selective Sample Replay (SSR)**, which mitigates the vanishing advantages problem by storing and prioritizing experiences with non-zero advantages, creating a more stable and curriculum-like learning process. Second, we introduce **Forced Rethinking**, which explicitly trains self-verification and correction capabilities by appending rethinking trigger tokens to training rollouts, encouraging the model to develop genuine self-reflection abilities without requiring teacher distillation.

Our approach is trained on approximately 38.9K carefully curated, verifiable multimodal reasoning queries spanning mathematics, science, and general reasoning tasks. Unlike previous approaches that apply rethinking uniformly, `VL-Rethinker` learns to rethink selectively, improving both accuracy and efficiency.

We evaluate `VL-Rethinker` across multiple challenging benchmarks including MathVista [Lu et al., 2023], MathVerse, MathVision, MMMU [Yue et al., 2024], MMMU-Pro, EMMA [Hao et al., 2025], and MEGA [Chen et al., 2024]. Our results demonstrate significant improvements over existing open-source models: our 72B model achieves 80.4% on MathVista testmini (+5.6 over open-source SoTA), while our 7B model substantially outperforms existing reasoning VLMs across all benchmarks.

Our main contributions are:

- We identify and formalize the vanishing advantages problem in GRPO for vision-language models, explaining training instability and shallow reasoning limitations.
- We propose Selective Sample Replay (SSR) to prioritize high-advantage experiences, yielding stable, curriculum-like reinforcement learning for multimodal reasoning.
- We introduce Forced Rethinking to explicitly train self-verification and correction capabilities without requiring teacher model distillation.
- We achieve new state-of-the-art results on multiple multimodal reasoning benchmarks at both 7B and 72B scales, demonstrating the effectiveness of our approach.

## 2 Preliminaries

### 2.1 Group Relative Policy Optimization (GRPO)

Group Relative Policy Optimization (GRPO) is a policy gradient method designed for training language models with binary rewards. Unlike traditional PPO [Schulman et al., 2017], which requires value function estimation, GRPO directly optimizes policies by comparing responses within groups.

Given a group of responses $\{y_1, y_2, ..., y_n\}$ to the same prompt $x$, GRPO computes relative advantages as:

$$A_i = r(x, y_i) - \frac{1}{n}\sum_{j=1}^{n} r(x, y_j) \tag{1}$$

where $r(x, y_i)$ is the reward for response $y_i$. The policy loss is then:

$$L_{\text{GRPO}} = -\frac{1}{n}\sum_{i=1}^{n} A_i \log \pi_\theta(y_i|x) \tag{2}$$

This approach is particularly suitable for binary reward settings common in reasoning tasks, where responses are either correct or incorrect.

### 2.2 The Vanishing Advantages Problem

While GRPO has shown promise in text-only settings, we identify a critical limitation when applied to vision-language reasoning: the *vanishing advantages problem*. As illustrated in Figure 2, this problem manifests in two ways:

**Reward Saturation:** As the model improves during training, an increasing proportion of query groups have all responses either correct (all rewards = 1) or incorrect (all rewards = 0). In both cases, the relative advantages become zero since $A_i = r_i - \bar{r} = 0$ when all responses have the same reward.

**Learning Signal Sparsity:** When advantages vanish, the gradient signal becomes zero: $\nabla_\theta L_{\text{GRPO}} = 0$. This effectively halts learning progress and prevents the model from developing deeper reasoning capabilities, as there are no longer distinguishable positive and negative examples within groups.

This problem is particularly acute in multimodal reasoning tasks, where the complexity of visual understanding combined with logical reasoning leads to training instability and prevents models from developing robust self-reflection capabilities.

### 2.3 Experience Replay in Reinforcement Learning

Experience replay, originally introduced for off-policy learning in deep Q-networks, has proven effective for improving sample efficiency and training stability. Prioritized Experience Replay (PER) [Schaul et al., 2015] further enhances this by sampling experiences based on their temporal difference error, focusing learning on the most informative examples.

However, standard experience replay approaches are not directly applicable to policy gradient methods like GRPO, which require on-policy data generation. Our approach adapts these principles specifically for the group-based policy optimization setting while maintaining the on-policy nature of GRPO.

## 3 VL-Rethinker: Our Method

We propose `VL-Rethinker`, which addresses the vanishing advantages problem through two complementary innovations: Selective Sample Replay (SSR) and Forced Rethinking. Figure 3 provides an overview of our method.

### 3.1 Selective Sample Replay (SSR)

To combat vanishing advantages, we introduce Selective Sample Replay (SSR), which maintains a replay buffer of experiences with non-zero advantages and uses prioritized sampling to focus learning on the most informative examples.

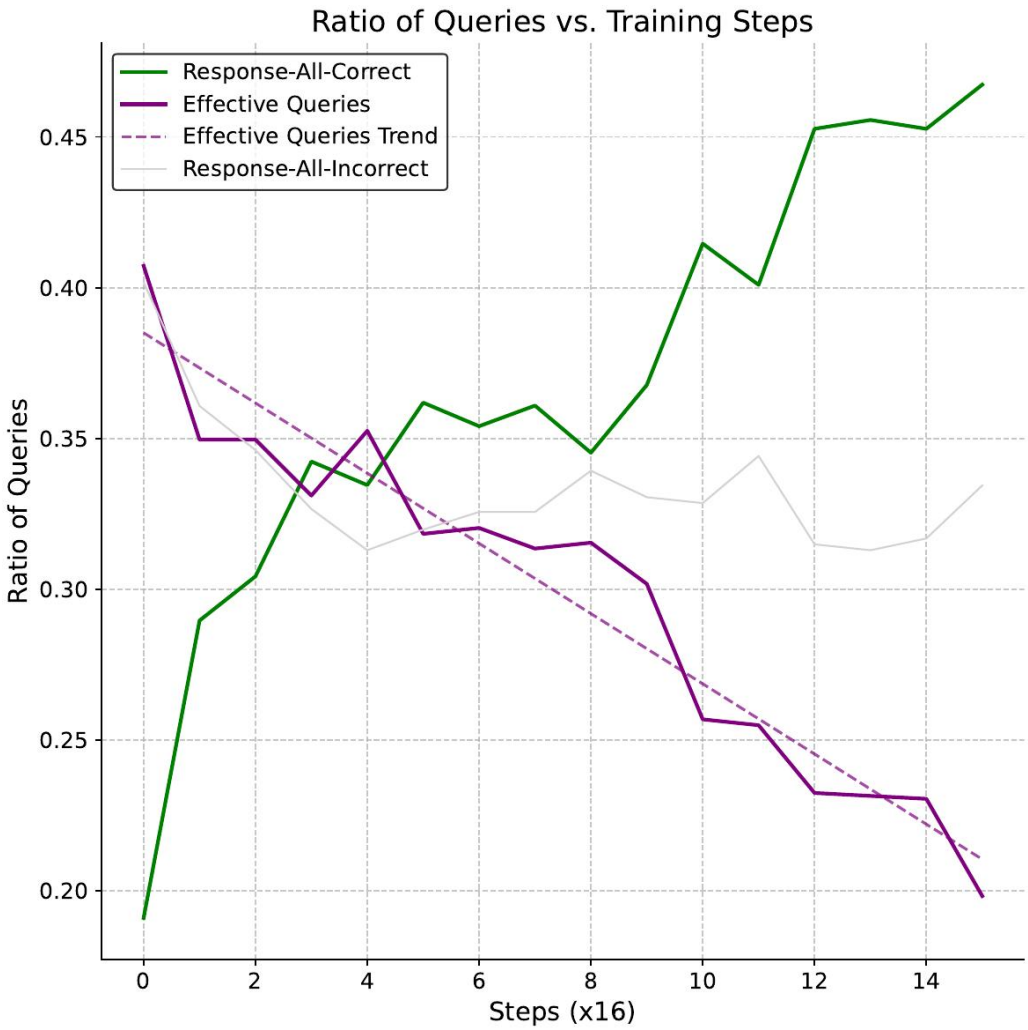


Figure 2: Illustration of the vanishing advantages problem in GRPO. As training progresses, the proportion of queries where all responses are correct (Response-All-Correct) increases while effective queries with mixed outcomes decrease, leading to sparse learning signals.

#### 3.1.1 Experience Storage and Filtering

SSR maintains a replay buffer $\mathcal{B}$ that stores experiences $(x, y, A)$ where $x$ is the input query, $y$ is the response, and $A$ is the advantage. We filter experiences based on their advantage values:

$$\mathcal{B}_t = \{(x, y, A) | |A| > \epsilon\} \cup \mathcal{B}_{t-1} \tag{3}$$

where $\epsilon$ is a small threshold (we use $\epsilon = 0.05$). This ensures that only experiences with meaningful advantage signals are retained for replay.

#### 3.1.2 Prioritized Sampling

During training, we sample from the replay buffer using a priority-based approach inspired by PER [Schaul et al., 2015], but adapted for advantage-based prioritization:

$$p(i) = \frac{|A_i|^\alpha}{\sum_j |A_j|^\alpha} \tag{4}$$

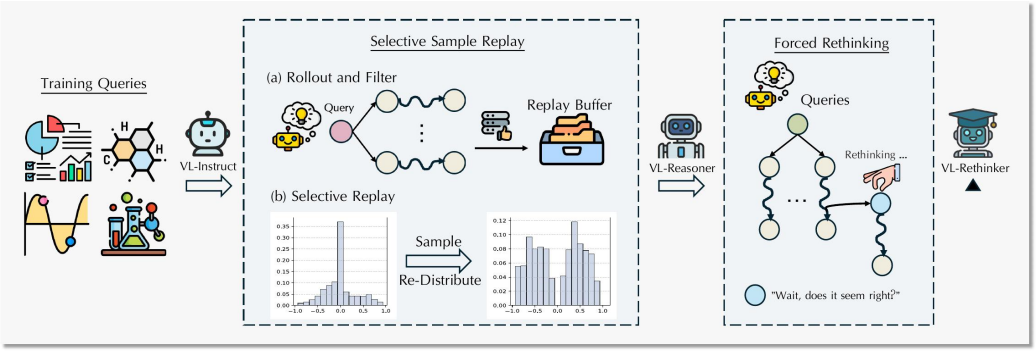


Figure 3: Method overview of `VL-Rethinker`. Our approach combines (1) Selective Sample Replay (SSR) to address vanishing advantages by prioritizing high-advantage experiences, and (2) Forced Rethinking to explicitly train self-verification capabilities through rethinking triggers.

where $\alpha$ controls the level of prioritization. Higher values of $\alpha$ concentrate sampling on experiences with larger advantages, creating a curriculum-like effect where the model focuses on the most challenging decision boundaries.

#### 3.1.3 Buffer Management

To maintain the near on-policy nature of GRPO while benefiting from experience replay, we implement a mixed sampling strategy:

$$\mathcal{D}_{\text{train}} = (1 - \lambda)\mathcal{D}_{\text{current}} + \lambda\mathcal{D}_{\text{replay}} \tag{5}$$

where $\mathcal{D}_{\text{current}}$ contains fresh samples from the current policy, $\mathcal{D}_{\text{replay}}$ contains samples from the replay buffer, and $\lambda$ controls the replay ratio. This ensures that learning remains primarily on-policy while benefiting from stable gradients provided by high-advantage historical examples.

### 3.2 Forced Rethinking

To develop genuine self-reflection capabilities without teacher distillation, we introduce Forced Rethinking, which explicitly trains the model to verify and correct its initial responses.

#### 3.2.1 Rethinking Triggers

During training rollouts, we append rethinking trigger tokens to a fraction $q$ of the generated responses:

$$y_{\text{rethink}} = y_{\text{initial}} \oplus \text{TRIGGER} \oplus y_{\text{revision}} \tag{6}$$

where $\oplus$ denotes concatenation, and TRIGGER is chosen from a set of self-reflection prompts:

- "Let me double-check this..."
- "Wait, let me reconsider..."
- "Actually, I should verify this step..."

#### 3.2.2 Training Objective

We combine the standard GRPO objective with an auxiliary supervised fine-tuning (SFT) loss for successful rethinking trajectories:

$$L_{\text{total}} = L_{\text{GRPO}} + \beta L_{\text{SFT}} \tag{7}$$

where:

$$L_{\text{SFT}} = -\sum_{(x,y)\in\mathcal{S}} \log \pi_\theta(y_{\text{rethink}}|x) \tag{8}$$

$$\mathcal{S} = \{(x,y)|r(x,y_{\text{rethink}}) > r(x,y_{\text{initial}})\} \tag{9}$$

This encourages the model to learn effective self-correction patterns while maintaining the RL signal for overall optimization.

#### 3.2.3 Selective Rethinking

Rather than applying rethinking uniformly, our approach learns to rethink selectively. During training, we apply rethinking triggers with probability $q$ that decreases over time:

$$q(t) = q_0 \cdot \exp(-t/\tau) \tag{10}$$

This schedule encourages the model to develop internal mechanisms for determining when rethinking is necessary, improving both accuracy and efficiency.

### 3.3 Algorithm

Algorithm 1 summarizes our complete training procedure. The key insight is that SSR provides stable gradients by focusing on high-advantage experiences, while Forced Rethinking develops the self-reflection capabilities necessary for complex multimodal reasoning.

**Algorithm 1** VL-Rethinker Training

```
Initialize policy π_θ, replay buffer B = ∅
for epoch t = 1, 2, ..., T do
    Generate responses {y_i} for batch of queries {x_j}
    Compute rewards {r_ij} and advantages {A_ij}
    Filter high-advantage experiences: E_t = {(x, y, A) | |A| > ε}
    Update replay buffer: B = B ∪ E_t
    for fraction q(t) of responses do
        Append rethinking trigger and generate revision
        if revision improves reward then
            Add to successful rethinking set S
        end if
    end for
    Sample replay data D_replay from B using prioritized sampling
    Combine current and replay data: D = (1 − λ)D_current + λD_replay
    Update policy: θ ← θ − ∇_θ(L_GRPO + βL_SFT)
end for
```

### 3.4 Advantages of Our Approach

Our method provides: (1) training stability through consistent learning signals; (2) curriculum learning via prioritized sampling; (3) distillation-free self-reflection development; and (4) improved efficiency through selective rethinking.

## 4 Experiments

### 4.1 Experimental Setup

#### 4.1.1 Datasets and Benchmarks

We evaluate `VL-Rethinker` on seven challenging multimodal reasoning benchmarks:

**Mathematical Reasoning:**

| Model | Math-Related | | | Multi-Discipline | | | Real-World |
|---|---|---|---|---|---|---|---|
| | MathVista testmini | MathVerse testmini | MathVision test | MMMU-Pro overall | MMMU val | EMMA full | MEGA core |
| General Vision-Language Models | | | | | | | |
| InternVL2-8B | 58.3 | - | 17.4 | 29.0 | 51.2 | 19.8 | 26.0 |
| InternVL2.5-8B | 64.4 | 39.5 | 19.7 | 34.3 | 56.0 | - | 30.4 |
| QwenVL2-7B | 58.2 | - | 16.3 | 30.5 | 54.1 | 20.2 | 34.8 |
| QwenVL2.5-7B | 68.2 | 46.3 | 25.1 | 36.9 | †54.3 | 21.5 | 35.0 |
| Llava-OV-7B | 63.2 | 26.2 | - | 24.1 | 48.8 | 18.3 | 22.9 |
| Kimi-VL-16B | 68.7 | 44.9 | 21.4 | - | †55.7 | - | - |
| Vision-Language Reasoning Models | | | | | | | |
| MM-Eureka-8B (Intern) | 67.1 | 40.4 | 22.2 | 27.8 | 49.2 | - | - |
| MM-Eureka-7B (Qwen) | 73.0 | 50.3 | 26.9 | - | - | - | - |
| R1-VL-7B | 63.5 | 40.0 | 24.7 | 7.8 | 44.5 | 8.3 | 29.9 |
| R1-Onevision-7B | 64.1 | 46.4 | 29.9 | 21.6 | - | 20.8 | 27.1 |
| OpenVLThinker-7B | 70.2 | 47.9 | 25.3 | 37.3 | 52.5 | 26.6 | 12.0 |
| `VL-Rethinker`-7B | **74.9** | **54.2** | **32.3** | **41.7** | **56.7** | **29.7** | **37.2** |
| Δ (Ours - Prev SoTA) | +4.7 | +6.3 | +2.4 | +4.4 | +0.7 | +3.1 | +2.2 |

Table 1: Comparison between our 7B model and other general and reasoning vision-language models. † means that the results are reproduced by us.

- **MathVista** [Lu et al., 2023]: 6,141 examples combining mathematical and visual reasoning tasks.
- **MathVerse**: Mathematical reasoning with visual contexts covering geometry, algebra, and statistics.
- **MathVision**: Challenging mathematical problems requiring deep visual understanding.

**Multi-Discipline Understanding:**

- **MMMU** [Yue et al., 2024]: 11.5K college-level questions across six core disciplines.
- **MMMU-Pro**: Enhanced version with more challenging multi-step reasoning requirements.
- **EMMA** [Hao et al., 2025]: Enhanced multimodal reasoning benchmark targeting organic multimodal reasoning.
- **MEGA** [Chen et al., 2024]: Large-scale benchmark with over 500 real-world tasks.

#### 4.1.2 Training Details

We train `VL-Rethinker` on approximately 38.9K curated, verifiable multimodal reasoning queries. Our training pipeline uses the following configuration:

- **Base Models:** We experiment with 7B and 72B parameter vision-language models.
- **SSR Parameters:** Replay ratio $\lambda = 0.3$, priority exponent $\alpha = 0.8$, advantage threshold $\epsilon = 0.05$.
- **Forced Rethinking:** Initial rethinking probability $q_0 = 0.4$, decay constant $\tau = 1000$, SFT weight $\beta = 0.1$.
- **Training:** 3 epochs with learning rate 1e-6, batch size 64, gradient accumulation steps 8.

### 4.2 Main Results

#### 4.2.1 7B Model Performance

Table 1 compares our 7B model with existing general and reasoning vision-language models. `VL-Rethinker`-7B achieves substantial improvements across all benchmarks:

- **Mathematical Reasoning:** 74.9% on MathVista testmini (+4.7 over previous SoTA), 54.2% on MathVerse testmini (+6.3), and 32.3% on MathVision test (+2.4).

| Model | Math-Related | | | Multi-Discipline | | | Real-World |
|---|---|---|---|---|---|---|---|
| | MathVista testmini | MathVerse testmini | MathVision test | MMMU-Pro overall | MMMU val | EMMA full | MEGA core |
| Proprietary Model | | | | | | | |
| OpenAI-o1 | 73.9 | 57.0 | 60.3 | 62.4 | 78.2 | 45.7 | 56.2 |
| OpenAI-GPT-4o | 60.0 | 41.2 | 30.6 | 51.9 | 69.1 | 32.7 | 52.7 |
| Claude-3.5-Sonnet | 67.7 | 47.8 | 33.5 | 51.5 | 68.3 | 35.1 | 52.3 |
| Gemini-2.0-Flash | 73.4 | 54.6 | 41.3 | 51.7 | 70.7 | 33.6 | 54.1 |
| Open-Source Models | | | | | | | |
| Llama4-Scout-109B | 70.7 | - | - | 52.2 | 69.4 | 24.6 | 31.8 |
| InternVL-2.5-78B | 72.3 | 51.7 | 34.9 | 48.6 | 61.8 | 27.1 | 44.1 |
| QvQ-72B | 71.4 | 48.6 | 35.9 | 51.5 | 70.3 | 32.0 | 8.8 |
| LLava-OV-72B | 67.5 | 39.1 | 30.1 | 31.0 | 56.8 | 23.8 | 29.7 |
| Qwen-2.5-VL-32B | 74.7 | 48.5 | 38.4 | 49.5 | †59.4 | 31.1 | 13.3 |
| Qwen-2.5-VL-72B | 74.8 | 57.2 | 38.1 | 51.6 | †67.0 | 34.1 | 49.0 |
| `VL-Rethinker-32B` | 78.8 | 56.9 | 40.5 | 50.6 | 65.6 | 37.9 | 19.9 |
| `VL-Rethinker-72B` | **80.4** | **63.5** | **44.9** | **55.9** | **68.8** | **38.5** | **51.3** |
| Δ (Ours - Open SoTA) | +5.6 | +6.3 | +6.8 | +3.7 | -1.4 | +4.4 | +2.3 |

Table 2: Comparison between our 72B model and other state-of-the-art models. The notation of † indicates reproduced results using our evaluation protocols.

| Model | RL-Algo | Data | MathVision | MathVista | MathVerse | MMMU-Pro | EMMA |
|---|---|---|---|---|---|---|---|
| `VL-Rethinker-7B` | SSR | 16K | 32.3 | 74.9 | 54.2 | 41.7 | 29.7 |
| w/o 'Forced-Rethinking' | SSR | 16K | 29.8 | 72.4 | 53.2 | 40.9 | 29.5 |
| - no SSR | Filter | 16K | 28.5 | 72.0 | 50.0 | 40.0 | 26.9 |
| - no SSR& Filter | GRPO | 16K | 26.0 | 70.9 | 51.4 | 38.8 | 26.2 |
| - no Text | SSR | 13K | 29.1 | 73.5 | 53.5 | 41.1 | 28.7 |
| - no Science&Text | SSR | 11K | 28.0 | 71.6 | 50.3 | 39.7 | 28.0 |

Table 3: Ablation Results to show the impact of SSR and Data Mix.

- **Multi-Discipline:** 41.7% on MMMU-Pro overall (+4.4), 56.7% on MMMU val (+0.7), and 29.7% on EMMA full (+3.1).
- **Real-World Tasks:** 37.2% on MEGA core (+2.2), demonstrating strong generalization to diverse real-world scenarios.

Notably, our 7B model substantially outperforms specialized reasoning models like R1-OneVision-7B and OpenVLThinker-7B, while also surpassing much larger general-purpose models.

#### 4.2.2 72B Model Performance

Table 2 shows that our 72B model achieves new state-of-the-art results among open-source models and competitive performance with proprietary systems:

- **Mathematical Benchmarks:** 80.4% on MathVista testmini (+5.6 over open-source SoTA), 63.5% on MathVerse testmini (+6.3), and 44.9% on MathVision test (+6.8).
- **Multi-Discipline Tasks:** 55.9% on MMMU-Pro overall (+3.7) and 38.5% on EMMA full (+4.4).
- **Comparison with Proprietary Models:** Our 72B model achieves competitive performance with OpenAI o1 on several benchmarks, demonstrating the effectiveness of our approach.

### 4.3 Ablation Studies

#### 4.3.1 Component Analysis

Table 3 presents comprehensive ablation studies analyzing the contribution of each component:

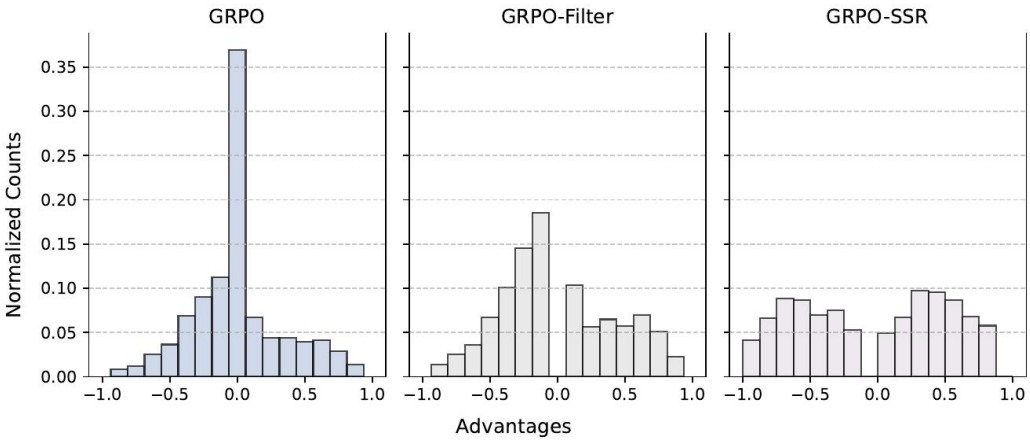


Figure 4: Advantage distribution comparison between GRPO, GRPO-Filter, and GRPO-SSR during training. SSR successfully mitigates the concentration of advantages at zero, providing more stable learning signals.

**Effect of SSR:** Removing SSR and using only filtering ("no SSR") leads to consistent performance drops across all benchmarks, with particularly notable decreases in MathVision (32.3→28.5) and EMMA (29.7→26.9).

**Effect of Forced Rethinking:** Removing the Forced Rethinking component ("w/o Forced-Rethinking") results in performance degradation, especially on complex reasoning tasks like MathVision (32.3→29.8).

**Vanilla GRPO Baseline:** Using standard GRPO without any modifications ("no SSR & Filter") shows the largest performance drops, confirming the importance of addressing the vanishing advantages problem.

#### 4.3.2 Data Mix Analysis

We also analyze the effect of different data compositions:

- **No Text:** Removing text-only reasoning data slightly decreases performance across most benchmarks.
- **No Science & Text:** Further removing science-related data leads to more substantial drops, particularly on MMMU-Pro and EMMA.

These results demonstrate that our curated mix of mathematical, scientific, and general reasoning data is crucial for optimal performance.

### 4.4 Training Dynamics Analysis

Figure 4 illustrates how SSR addresses the vanishing advantages problem. Standard GRPO shows a sharp concentration of advantages around zero, indicating reward saturation. GRPO-Filter partially addresses this but still shows significant concentration at zero. In contrast, GRPO-SSR (our approach) maintains a more balanced advantage distribution throughout training, providing consistent learning signals.

#### 4.4.1 Convergence Analysis

Our training curves show that SSR leads to more stable convergence compared to standard GRPO. The prioritized replay mechanism prevents training from stalling when reward saturation occurs, allowing the model to continue improving its reasoning capabilities.

### 4.5 Qualitative Analysis

We observe several important qualitative improvements in `VL-Rethinker`:

**Self-Correction:** The model demonstrates improved ability to catch and correct its own mistakes, particularly in mathematical calculations and logical reasoning steps.

**Uncertainty Awareness:** Through Forced Rethinking training, the model learns to express uncertainty appropriately and seek verification when confidence is low.

**Reasoning Depth:** Compared to baseline models, `VL-Rethinker` produces more thorough and systematic reasoning chains, especially for complex multi-step problems.

### 4.6 Computational Efficiency

Despite the additional complexity of SSR and Forced Rethinking, our approach maintains reasonable computational overhead:

- **Training Overhead:** SSR adds approximately 15% to training time due to replay buffer management.
- **Inference Efficiency:** Selective rethinking reduces unnecessary computation compared to uniform rethinking strategies.
- **Memory Usage:** The replay buffer requires additional memory but scales linearly with buffer size, which can be tuned based on available resources.

## 5 Related Work

### 5.1 Vision-Language Reasoning

Recent advances in vision-language models have demonstrated impressive capabilities, but complex reasoning tasks requiring deep visual-textual integration remain challenging. MathVista [Lu et al., 2023] and MMMU [Yue et al., 2024] have highlighted significant gaps in mathematical and scientific reasoning.

Most approaches focus on supervised learning paradigms that may not effectively teach models to verify and correct their reasoning.

### 5.2 Reinforcement Learning for Language Models

Reinforcement learning has shown promise for improving language models, particularly through RLHF approaches. PPO [Schulman et al., 2017] has been widely adopted but requires unstable and expensive value function estimation.

GRPO emerged as an alternative using within-group comparisons. However, as we demonstrate, GRPO suffers from vanishing advantages in multimodal reasoning settings.

### 5.3 Self-Reflection and Verification in AI

Self-reflection concepts gained attention following GPT-o1's success. Traditional approaches rely on external verification or teacher distillation [Lightman et al., 2023], which can be resource-intensive.

Recent work explores training models to generate verification capabilities, but most focus on text-only domains using supervised fine-tuning that may not generalize well.

### 5.4 Experience Replay in Reinforcement Learning

Experience replay has been a cornerstone of deep RL since DQN. Prioritized Experience Replay [Schaul et al., 2015] demonstrated improved sample efficiency through importance sampling. Our SSR approach adapts these principles to policy gradient methods while maintaining on-policy requirements.

### 5.5 Chain-of-Thought and Step-by-Step Reasoning

Chain-of-thought prompting [Wei et al., 2022] effectively elicits reasoning but focuses on prompting rather than training intrinsic capabilities. Step-by-step verification [Lightman et al., 2023] showed process-level supervision effectiveness. Our Forced Rethinking applies these insights directly during RL training.

### 5.6 Multimodal Reasoning Models

Recent efforts like R1-VL [Zhang et al., 2025] improved multimodal reasoning but face training instability and difficulty developing self-reflection without extensive supervision. Our work provides a principled solution to vanishing advantages while training self-reflection capabilities through forced rethinking.

## 6 Conclusion

In this work, we addressed a critical gap in multimodal reasoning capabilities by identifying and solving the vanishing advantages problem in Group Relative Policy Optimization for vision-language models. Our proposed approach, `VL-Rethinker`, combines two key innovations: Selective Sample Replay (SSR) and Forced Rethinking, which together enable stable training of sophisticated self-reflection capabilities in multimodal settings.

### 6.1 Summary of Contributions

Our main contributions include: (1) identifying and formalizing the vanishing advantages problem in GRPO for multimodal reasoning; (2) proposing SSR to maintain training stability through prioritized experience replay; (3) introducing Forced Rethinking for self-verification without teacher distillation; and (4) achieving state-of-the-art results on multiple benchmarks at 7B and 72B scales.

### 6.2 Impact and Significance

Our work contributes: (1) SSR as a general framework for stabilizing policy gradient training with sparse rewards; (2) competitive performance with larger proprietary models using reasonable resources; and (3) more stable and efficient training through addressing vanishing advantages.

### 6.3 Limitations

Limitations include: (1) SSR introduces computational and memory overhead; (2) hyperparameter sensitivity requiring domain-specific tuning; and (3) evaluation focus on mathematical/scientific reasoning limits generalizability claims.

### 6.4 Future Directions

Future directions include: (1) adaptive rethinking through uncertainty estimation; (2) multi-scale reasoning across abstraction levels; (3) domain transfer beyond mathematical reasoning; and (4) theoretical analysis of SSR benefits.

### 6.5 Broader Impact

Capable multimodal reasoning systems offer significant societal benefits including enhanced education, research assistance, and accessibility. However, careful consideration of potential misuse and safety measures remains important. Our focus on genuine self-reflection over pattern matching contributes to more reliable and trustworthy AI systems.

In conclusion, `VL-Rethinker` significantly advances multimodal reasoning training through RL by addressing policy optimization challenges and developing novel self-reflection mechanisms. Strong empirical results demonstrate practical value for advancing multimodal AI systems.

## References


Jiacheng Chen, Tianhao Liang, Sherman Siu, Zhengqing Wang, Kai Wang, Yubo Wang, Yuansheng Ni, Wang Zhu, Ziyan Jiang, Bohan Lyu, et al. Mega-bench: Scaling multimodal evaluation to over 500 real-world tasks. *arXiv preprint arXiv:2410.10563*, 2024.

Yunzhuo Hao, Jiawei Gu, Huichen Will Wang, Linjie Li, Zhengyuan Yang, Lijuan Wang, and Yu Cheng. Can mllms reason in multimodality? emma: An enhanced multimodal reasoning benchmark. *arXiv preprint arXiv:2501.05444*, 2025.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*, 2023.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.

Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. *arXiv preprint arXiv:1511.05952*, 2015.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024.

Jingyi Zhang, Jiaxing Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. R1-vl: Learning to reason with multimodal large language models via step-wise group relative policy optimization. *arXiv preprint arXiv:2503.12937*, 2025.